

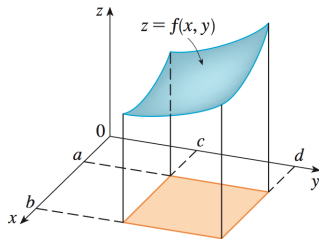
Topic: Gradient

I. Scalar Functions

Defn: A Scalar Function is a function

Ex: $f(x) = x^2$ is a scalar function since
 $x \in \mathbb{R}$ and $f(x) \in \mathbb{R}$

Ex: $f(x, y) = x - x\sqrt{y}$ is a scalar function since
 $(x, y) \in \mathbb{R}^2$ and $f(x, y) \in \mathbb{R}$



Ex: $f(x, y, z) = \frac{\cos(z)}{x} - x$ is a scalar function since
 $(x, y, z) \in \mathbb{R}^3$ and $f(x, y, z) \in \mathbb{R}$

II. Gradient

Defn: Let D_f be the domain of the scalar function $f(x, y)$ and D_g be the domain of the scalar function $g(x, y, z)$. If f is differentiable on D_f and g is differentiable on D_g then the **gradient** of $f(x, y)$ and $g(x, y, z)$ is given by

$$\nabla f =$$

$$\nabla g =$$

Ex: $f(x, y) = xy + y$. Find ∇f

Ex: $f(x, y, z) = z \ln(xy^2 + y)$. Find ∇f at the point $P(0, 1, 2)$

Prove the following two properties of gradients:

$$\text{Recall: } \nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right]$$

1. $\nabla(f + g) = \nabla f + \nabla g$

Proof : $\nabla(f + g) =$

2. $\nabla(f \cdot g) = \nabla f \cdot g + f \cdot \nabla g$

Proof : $\nabla(fg) =$

III. Additional Concepts

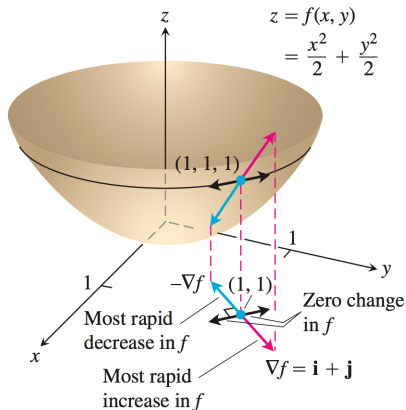
Thm: $\nabla f(P)$ points in the direction of

Defn: The **Directional Derivative** of a function f , $D_{\mathbf{u}}f$, in the direction of the unit vector $\mathbf{u}$ is

$$D_{\mathbf{u}}f = \quad \text{so we also have} \quad D_{\mathbf{u}}f =$$

Ex: Let $f(x, y) = \frac{x^2}{2} + \frac{y^2}{2}$, $P(1, 1)$ & $\mathbf{u} = \left[\frac{1}{2}, \frac{\sqrt{3}}{2} \right]$

Compute $D_{\mathbf{u}}f(P)$



Ex: Is there a direction $\mathbf{u}$ in which the rate of change of the temperature function $T(x, y) = x^2 - 3xy + 4y^2$ at $P(1, 2)$ equals $14^\circ\text{C}/\text{ft}$? Give reasons for your answer.

ANSWER:

REASON: